Concentrated nonnegative repairs: a consolidated appendix to the flat-lattice rank dossier

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Abstract

This appendix assembles one actual finite-flat mechanism: a repair supported at a single lattice site has arbitrarily prescribed nonnegative directional mean, yet vanishes on the kernel of a fixed-dimensional origin-difference map. A constructed divergence-free low-energy band then forces observation rank of volume order. We state the explicit integer-tail inequality and its uniform bounded-rank consequence, identify their exact Lean declarations, and preserve the failed and successful reproduction records. This is an appendix for the existing coarse-information dossier, not a separate paper or deposit. The mechanism and its quantifier closure are classical; no independent novelty, interacting gauge-theory gap or continuum limit is claimed.

1 Objects and scope

Let $D = d + 2 \geq 2$, $n \geq 1$, and $N_c \geq 2$. Work on the periodic unit-spacing lattice $X_n = (\mathbb{Z}/n\mathbb{Z})^D$ with the actual real physical one-cochain space used by the formalization. Its coefficient space G has real dimension $g = N_c^2 - 1$. Write $A_j(x) \in G$ for a field component and $\Delta_i A_j(x) = A_j(x + e_i) - A_j(x)$ for the forward difference. All norms and energies below use the repository's actual finite-flat normalization. Write $\mathcal{E}_0(A)$ for its flat gauge Hodge form and $\mathcal{E}_w(A)$ for its componentwise weighted repair form. The gauge condition is the actual flat divergence constraint associated with the stated adjoint model ρ.

For $x_0 \in X_n$ and $b \in \mathbb{R}$, define

$$w_{i,x,j} = b\,\mathbf{1}_{\{x=x_0\}}, \qquad (\mathcal{J}_{x_0} A)_{ij} = \Delta_i A_j(x_0).$$

The parameters i, j both run over D directions. Thus the codomain of $\mathcal{J}_{x_0}$ has dimension $D^2 g$, independently of n and b. This is the earlier site-gradient map in a different finite-product packaging, not an independently introduced observation mechanism.

Lemma 1 (Actual energy and exact mean)**.** *For the actual concentrated weight,*

$$\mathcal{E}_w(A) = b \sum_{i,j} \|\Delta_i A_j(x_0)\|^2.$$

In particular $\mathcal{E}_w = 0$ on $\ker \mathcal{J}_{x_0}$, for every real b. If $m \geq 1$ and $b = (m-1)n^D$, then $w \geq 0$ and, for every i, j,

$$\sum_{x \in X_n} \max(1 + w_{i,x,j}, 0) = n^D m.$$

The last identity is an equality of actual directional sums, not a budget certificate supplied as an extra hypothesis. It follows from the support definition and the already derived repair identity. In particular $m = 1$ gives the zero repair.

2 Original local nonlinear data

An observation is a map C from the actual physical cochains to a real finite-dimensional normed space F. Assume $C(0) = 0$ and differentiability at zero with derivative $L = DC(0)$. Let $\Phi : F \to \mathbb{R}$ obey

$$\Phi(y) \le K\|y\|^2 \quad (\|y\| < \delta), \qquad \delta > 0.$$

No differentiability of Φ is assumed. The local coercivity assertion to be tested is, for some $c > 0$ and $r > 0$,

$$c\|A\|^2 \le \mathcal{E}_0(A) + \mathcal{E}_w(A) + \Phi(C(A)) \quad \text{if } \operatorname{div}_\rho A = 0 \text{ and } \|A\| < r. \tag{1}$$

These are the original local nonlinear assumptions. No rank certificate, band-dimension certificate, supplied injection or compressed-energy certificate replaces them. In a family indexed by n, the observation, K, δ and r may vary with n.

3 One mechanism, from a finite band to the integer tail

The actual constructed positive coexact physical band of cutoff R has dimension

$$H_R = 2R(2R+1)^{D-1},$$

lies in the actual divergence kernel, and satisfies

$$\mathcal{E}_0(A) \le \beta_{n,R}\|A\|^2, \qquad \beta_{n,R} = \frac{4\pi^2 D R^2}{n^2}.$$

Its existence, dimension, divergence vanishing and energy bound are earlier derived facts of the actual formal construction, not hypotheses of the final rank theorem.

Theorem 1 (Actual finite-band obstruction)**.** *Suppose $2R < n$ and $\beta_{n,R} < c$. If (1) holds with the concentrated repair, then*

$$H_R \le \dim_{\mathbb{R}} F + D^2(N_c^2 - 1).$$

The amplitude b can be any real number.

Proof. Restrict $(L, \mathcal{J}_{x_0})$ to the actual coexact band. An element of its kernel has zero repair energy and is invisible to the derivative of C. Along tA, differentiability gives $C(tA) = o(t)$; the upper-quadratic penalty therefore contributes at most $o(t^2)$. For sufficiently small t, the original neighbourhood conditions apply. Dividing the local inequality by t^2 gives $c\|A\|^2 \le \mathcal{E}_0(A) \le \beta_{n,R}\|A\|^2$ in the limit. The strict gap forces $A = 0$. The restricted combined map is injective, so rank-nullity and the actual codomain dimension yield the assertion. □

Choose $k \ge 3$ so that

$$\frac{4\pi^2 D}{k^2} \le \frac{c}{2}. \tag{2}$$

Such an integer exists for every $c > 0$. For every integer $n \geq k$, put $R = \lfloor n/k \rfloor$. The quotient estimates derive

$$0 < R, \quad 2R < n, \quad \beta_{n,R} \leq \frac{4\pi^2 D}{k^2} < c, \quad n^D \leq k^D H_R.$$

The last comparison is a natural-number inequality; it is not restricted to a divisibility subsequence.

Theorem 2 (Full integer-tail rank and actual nonnegative profile)**.** *Under the original local data, for $n \geq k$ and* (2)*,*

$$n^D \leq k^D(\dim_{\mathbb{R}} F + D^2(N_c^2 - 1)). \tag{3}$$

For every $m \geq 1$, the actual choice $b = (m-1)n^D$ is nonnegative and has the exact directional positive-total mean m stated in Lemma 1; the same rank bound applies if its original local coercivity assertion holds.

Proof. Combine the derived quotient-volume comparison with Theorem 1. The profile statements are the already proved actual energy/mean adapters. □

The finite additive cost $D^2(N_c^2 - 1)$ must be retained. Neither this constant nor the chosen cutoff is asserted optimal.

4 Uniform bounded-rank consequence

Theorem 3 (Explicit uniform tail)**.** *Fix integers $d \geq 0$ and $N_c \geq 2$, a positive real c, and a natural rank budget q. Choose k by* (2)*, and set*

$$N = \max\{k,\ k^D(q + D^2(N_c^2 - 1)) + 1\}.$$

For every $n \geq N$, every $m \geq 1$, every $x_0 \in X_n$ and every observation space of dimension at most q, the original local assertion (1) *is false for the actual profile $w_{i,x,j} = (m-1)n^D \mathbf{1}_{\{x=x_0\}}$ and the stated local data. The threshold is independent of m, C, K, δ and r.*

Proof. If the assertion held, (3) and $\dim F \leq q$ would imply $n^D \leq k^D(q + D^2(N_c^2 - 1)) < n$. But $n \geq k \geq 3$ and $D \geq 2$ give $n \leq n^D$, a contradiction. The formal growth helper uses ordered power monotonicity. □

The quantifiers permit local radii and penalty constants to change at every n. A family with growing observation rank, or a coercivity constant tending to zero, is not excluded. The underlying arbitrary-amplitude obstruction is stronger than the intended nonnegative specialization; the premise $m \geq 1$ identifies that class but is not used in the rank contradiction itself.

5 Relation to the mean-budget question

The relevant distinction is between an infimum over profiles and a statement uniform over every profile with the same mean. The example above rules out the latter form of bounded-rank saving. It does not refute the earlier infimum-over-profiles scaling argument: a spatially constant repair and a concentrated repair have different distributions despite equal means. The block-mean upper construction and its tensorized variance estimate remain analytic in the current dossier; they are

not promoted here to Lean endpoints. This appendix consolidates the actual lower-obstruction chain.

Fourier localization, finite-rank perturbations and rank-nullity are classical. The uniform theorem is an elementary logical consequence of the preceding actual volume inequality. The primary-source dependency comparison establishes that it is not an independent new mechanism. No exhaustive external priority search or new general spectral principle is claimed. Relevance is limited to a precise, reproducible obstruction to the proposed mean-only repair route in this finite-flat model.

6 Formal coverage and reproduction record

The declaration map accompanying this PDF supplies full declaration names, source revisions, exact code paths, reproduction runs and output hashes. The exact primary code revision is retained independently of the manuscript and its assessment. The grouped stages are:

1. Actual support/energy annihilation: proof `fce61c6f`, runs 100/100b; the earlier failed 099 pair is retained in the source dossier.

2. Actual site-gradient identity, repair energy and exact directional budget: the same proof revision, runs 101/101b.

3. Actual coexact combined-map injection and finite-band rank: proof `f9332b26`, runs 102/102b.

4. Scalar cutoff, integer-tail volume rank and actual nonnegative profile: proof `82f7e201`, runs 103/103b.

5. Explicit bounded-rank tail: corrected proof `9e6298ef`, runs 105/105b. The initial proof `3281eb95` failed both 104 runs at an auxiliary power rewrite; full failed reports and the correction are preserved.

Every successful pair used fresh own Colab Linux clones at one exact source revision, the pinned Lean/Mathlib toolchain, compiler exit records and raw axiom reports. Matching output hashes are preparing-agent reproduction evidence, not independent scientific review. The current family pair has three exact oracle reports, ten source hashes and three real evidence-checker mutations rejected in both normal and optimized Python. Only `propext`, `Classical.choice` and `Quot.sound` occur in its reported axiom lists. No Windows Lean or Lake compilation was performed for these units.

The public-candidate package is assembled from an explicit allowlist of mathematical sources, reports, validators, this appendix and its map. Historical access-state snapshots and credentials are outside it. Its assembly and hash checks do not constitute a new Lean run. The package still requires ordinary artifact-level publication permissions; it is not an AIRR receipt or acceptance.

7 Model assessments and review context

The first identified preparing-agent assessment of the exact PDF belongs in a separate complete report, recorded against its SHA256 after all rendered pages have been visually inspected. This section reports no numerical PDF score and does not pre-approve the artifact. Source-level

first assessments remain separate: the energy and mean adapters received 5/10; finite-band and volume-rank sources received 6/10; the failed family source and the corrected family source each have their own first 5/10 report and seven objections. The corrected source was read and assessed anew; its score was not inherited from the failed one. These scores are not a score for this PDF.

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The previously frozen 21-page consolidated manuscript, SHA256 beginning `97bdcba4`, remains unchanged with its own first exact-PDF assessment and pending artifact-specific publication question. This appendix neither inherits that score nor replaces that permission request. It is not a new deposit, an accepted article, or evidence for a physical mass gap. PR29 remains 20/41 and TermSource remains zero; the interacting and continuum problems remain open.